

E-Commerce Sales & Customer Analytics

Data Dictionary & SQL Analysis Guide

Document ID	EC-BA-006
Version	1.0
Data Period	January 2023 - March 2023
Dataset	300 illustrative order records / 120 possible customer IDs

Portfolio note: Simulated data created for BA portfolio demonstration; not production/company data.

1. Dataset Purpose

The dataset supports sales, customer, product, regional and returns analysis and provides a practical source for SQL, Excel and Power BI exercises.

2. Data Dictionary

Field	Type	Description	Business Rule
order_id	Text	Unique order identifier.	One row represents one order record.
order_date	Date	Order transaction date.	Used for trend and period analysis.
customer_id	Text	Customer identifier.	Used for customer-level analysis.
region	Text	Sales region.	Five illustrative regions.
category	Text	Product category.	Electronics, Home, Fashion, Beauty.
product	Text	Product name.	Product within category.
quantity	Integer	Units in the order.	Must be greater than zero.
unit_price	Decimal	Price per unit before discount.	Illustrative currency amount.
discount_pct	Decimal	Discount percentage applied.	Expected range 0–100.
revenue	Decimal	Net order revenue after discount.	Cancelled orders have zero revenue.
order_status	Text	Completed, Returned or Cancelled.	Used to determine qualifying metrics.
returned_flag	Integer	Return indicator: 1 or 0.	1 when status is Returned.

3. KPI Calculation Logic

KPI	Calculation
Total Revenue	SUM(revenue) for qualifying reporting records.
Completed Orders	COUNT of orders where order_status = Completed.
Average Order Value	Total Revenue / Completed Orders.
Return Rate	Returned orders / qualifying non-cancelled orders × 100.
Revenue Share	Category revenue / total revenue × 100.
Repeat Customer	Customer with more than one qualifying completed order under the portfolio rule.

4. SQL Analysis Covered

The accompanying SQL file contains queries for overall KPIs, monthly trends, regional performance, category contribution, top products, returns, new versus repeat customers, high-value customers, region/category analysis and data-quality checks.

5. BA Analysis Workflow

Business question → KPI definition → data profiling → SQL analysis → validate results → visualize in Excel/Power BI → derive insight → recommend business action.

6. Interview Talking Point

Explain that SQL was used not just to retrieve data, but to validate business definitions, segment performance, identify exceptions and provide evidence for recommendations. Always confirm KPI rules with business stakeholders before presenting results.